

SURROGATE MODELING OF A BINARY DISTILLATION COLUMN USING DWSIM AND MACHINE LEARNING

Screening Task 3 — Benzene-Toluene Binary Distillation

Technical Illustration

Benzene-Toluene Distillation Process Flow

Benzene + Toluene Feed → *Distillation Column* → *Distillate + Bottoms*

Technologies: DWSIM | Peng-Robinson | Machine Learning | XGBoost

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Executive Summary

147 Unique Successful Simulations

Rigorous simulations performed using DWSIM to generate a comprehensive dataset.

7 Input Variables

Key operational parameters defining the distillation process.

4 Target Variables

Crucial output metrics including product compositions and energy duties.

0.9388 Average Test R^2 — XGBoost

XGBoost model demonstrates superior predictive accuracy across all target variables.

This report details the development and evaluation of machine learning surrogate models for a benzene-toluene binary distillation column. Leveraging 147 unique successful simulations generated in DWSIM, four machine learning approaches—Linear Regression, Random Forest, XGBoost, and Artificial Neural Networks (ANN)—were rigorously compared. The XGBoost model emerged as the optimal choice, achieving an average test R^2 of 0.9388, demonstrating its capability to rapidly and accurately predict key distillation outcomes.

Problem Statement

Objective

The primary objective of this project is to develop a data-driven surrogate model that can rapidly predict the performance of a benzene-toluene binary distillation column. This surrogate will serve as

a fast and efficient alternative to computationally intensive physics-based simulations for process optimization and real-time analysis.

Workflow

- **DWSIM Simulation:** Conducted 147 unique, successful steady-state simulations of the distillation column using DWSIM.
- **Dataset Generation:** Compiled simulation results into a structured dataset.
- **Machine Learning Model Development:** Explored and trained four different machine learning models.
- **Prediction & Evaluation:** Assessed model performance using various metrics and selected the best-performing surrogate.

Key Variables Defined

- **xD (Benzene mole fraction in distillate):** Purity of benzene in the top product.
- **xB (Benzene mole fraction in bottoms):** Purity of benzene in the bottom product.
- **QC (Condenser Duty):** Energy required for condensation in kW.
- **QR (Reboiler Duty):** Energy required for vaporization in kW.

DWSIM Model Configuration

System Definition

The process under consideration is the binary distillation of a **Benzene–Toluene** mixture. This system was chosen for its well-understood thermodynamic properties and relevance in chemical engineering.

Thermodynamic Property Package

The **Peng–Robinson (PR)** equation of state was employed as the thermodynamic property package. The PR model is known for its accuracy in predicting vapor-liquid equilibrium (VLE) for a wide range of hydrocarbon systems, making it suitable for this binary mixture.

Process Flow Diagram

DWSIM Distillation Column Process Flow

The diagram illustrates the flow of materials and energy through the distillation column, including the feed stream, the column internals, the condenser, the reboiler, and the distillate and bottoms product streams.

Components:

1. **Feed:** Benzene–Toluene mixture entering the column.
2. **Distillation Column:** Featuring trays for vapor-liquid contact.

3. **Condenser:** Cools and liquefies the overhead vapor.
4. **Reboiler:** Vaporizes the bottom liquid.
5. **Distillate:** Purified benzene product withdrawn from the top.
6. **Bottoms:** Purified toluene product withdrawn from the bottom.

Dataset Generation

Data Acquisition Strategy

A systematic approach was used to generate a robust dataset covering a wide operational envelope. This involved simulating the distillation column under various conditions using DWSIM.

- **Initial Source Datasets:** Multiple simulation runs were initiated based on preliminary operating parameter ranges.
- **Successful Simulations:** A significant number of these simulations resulted in converged and physically meaningful steady-state conditions.
- **Data Filtering & Cleaning:** Runs that failed to converge or produced unrealistic outputs were discarded.
- **Duplicate Removal:** To ensure unique operating points and avoid model bias, duplicate simulations (based on input variables) were identified and removed.
- **Final Dataset:** The resulting dataset comprises 147 unique, successful operating points.

Feed Condition

For all simulations, the **feed molar flow rate was kept constant** at **10.00057 mol/s** to isolate the impact of other operating variables on the distillation performance.

Dataset Structure

Variable Type	Variable Name	Description
Input	Feed_T_K	Feed Temperature (Kelvin)
Input	Feed_P_Pa	Feed Pressure (Pascals)
Input	Feed_xBenzene	Feed Benzene Mole Fraction
Input	Number_of_Stages	Total Number of Trays in Column
Input	Feed_Stage	Tray Number where Feed is Introduced
Input	Reflux_Ratio	Ratio of liquid returned to column to product withdrawn
Input	Bottoms_Flow_mol_s	Bottoms Product Molar Flow Rate (mol/s)
Output	xD_Benzene	Distillate Benzene Mole Fraction
Output	xB_Benzene	Bottoms Benzene Mole Fraction
Output	QC_kW	Condenser Duty (Kilowatts)
Output	QR_kW	Reboiler Duty (Kilowatts)

Operating Domain

Parameter	Unit	Minimum	Maximum	Distribution Notes
Feed Temperature	K	345.0	384.9	
Feed Pressure	Pa	9.0×10^4	1.1×10^5	
Feed Benzene Mole Fraction		0.30	0.70	
Number of Stages		10	12	85 simulations with 10 stages, 62 with 12 stages.
Feed Stage		2	9	
Reflux Ratio		1.509	5.0	
Bottoms Flow Rate	mol/s	4	6	
Distillate Benzene Mole Fraction (xD)		0.5374	0.9993	
Bottoms Benzene Mole Fraction (xB)		0.001849	0.4441	
Condenser Duty (QC)	kW	0.3341	1.105	
Reboiler Duty (QR)	kW	0.05296	1.121	

Data Quality and Exploratory Data Analysis (EDA)

Data Preprocessing

Before model training, the dataset was split into training and testing sets using an 80/20 ratio. A was utilized to ensure reproducibility of the split.

Exploratory Data Analysis

A correlation heatmap was generated to visualize the relationships between input and output variables. This analysis helps in understanding which process parameters have the most significant impact on the distillation outcomes.

Correlation Heatmap

Key Observations from Correlation Analysis:

- Feed Benzene Composition (Feed_xBenzene):** Shows strong negative correlation with and positive correlation with , as expected.
- Reflux Ratio (Reflux_Ratio):** Exhibits a strong positive correlation with and negative correlation with , indicating its critical role in separation efficiency.
- Energy Duties (QC, QR):** Both condenser and reboiler duties are significantly influenced by the reflux ratio and the separation achieved (indicated by xD and xB).

Machine Learning Methodology

Model Exploration

Four distinct machine learning models were selected and trained to serve as potential surrogates for the distillation column:

- Linear Regression:** A fundamental linear model assuming a linear relationship between inputs and outputs.
- Random Forest:** An ensemble method using multiple decision trees to improve robustness and accuracy.
- XGBoost (Extreme Gradient Boosting):** A highly optimized gradient boosting algorithm known for its performance and speed.
- Artificial Neural Network (ANN):** A deep learning model with a flexible architecture capable of learning complex non-linear relationships.

Model Specifics

- ANN Architecture:** The ANN comprised an input layer, two hidden layers (64 and 32 neurons respectively) with ReLU activation functions, and an output layer. The Adam optimizer was used, with early stopping implemented to prevent overfitting.
- XGBoost Hyperparameters:** Key parameters for XGBoost were set as follows: and , with a .

Workflow Pipeline

ML Pipeline Diagram

The diagram outlines the sequence of operations: raw dataset is subjected to preprocessing, followed by a train-test split. Four distinct ML models are then trained and evaluated. Cross-validation is performed on the best-performing models for robust performance assessment.

Model Performance Comparison

Performance Metrics

The performance of each model was evaluated using standard regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). A higher R^2 indicates a better fit to the data.

Performance Table

Model	Target	MAE	RMSE	R²
Linear Regression	xD_Benzene	0.0433	0.0501	0.8684
Linear Regression	xB_Benzene	0.0442	0.0508	0.8060
Linear Regression	QC_kW	0.0162	0.0209	0.9856
Linear Regression	QR_kW	0.0571	0.0673	0.9359
Random Forest	xD_Benzene	0.0322	0.0477	0.8807
Random Forest	xB_Benzene	0.0346	0.0484	0.8244
Random Forest	QC_kW	0.0401	0.0506	0.9161
Random Forest	QR_kW	0.0744	0.1134	0.8182
XGBoost	xD_Benzene	0.0282	0.0387	0.9214
XGBoost	xB_Benzene	0.0188	0.0245	0.9548
XGBoost	QC_kW	0.0160	0.0200	0.9869
XGBoost	QR_kW	0.0433	0.0772	0.9157
ANN	xD_Benzene	0.0456	0.0596	0.8138
ANN	xB_Benzene	0.0591	0.0905	0.3845
ANN	QC_kW	0.0458	0.0587	0.8870
ANN	QR_kW	0.0562	0.0688	0.9330

Performance Comparison Charts

Performance Charts

Visualizations comparing R^2 , MAE, and RMSE across the four models for each target variable. XGBoost consistently demonstrates superior performance, particularly for QC_kW and $xB_Benzene$.

Cross-Validation Results

Robustness Assessment

To ensure the generalization capability of the models, 5-fold cross-validation was performed. This technique divides the training data into five subsets, training the model on four subsets and validating on the remaining one, repeating this process five times.

Cross-Validation Table (XGBoost)

The following table summarizes the 5-fold cross-validation results for the XGBoost model, providing mean and standard deviation for key metrics:

Target	Mean R ²	Std R ²	Mean MAE	Mean RMSE
xD_Benzene	0.9091	0.0382	0.0296	0.0397
xB_Benzene	0.9098	0.0138	0.0264	0.0359
QC_kW	0.9739	0.0150	0.0202	0.0271
QR_kW	0.9399	0.0440	0.0386	0.0572

Feature Importance Analysis

Identifying Key Drivers

Feature importance analysis was performed on the trained XGBoost model to identify which input variables have the most significant impact on the prediction of output variables. This provides valuable engineering insights into the distillation process.

Feature Importance Chart

XGBoost Feature Importance

The horizontal bar chart displays the relative importance of each input feature in the XGBoost model. Features are ranked from most to least influential.

Most Influential Features:

- Reflux Ratio:** Consistently identified as the most critical factor, significantly affecting separation efficiency and product purity.
- Feed Temperature:** Plays a substantial role, influencing the thermodynamic state of the feed entering the column.
- Feed Benzene Composition:** Directly impacts the achievable separation targets.
- Bottoms Withdrawal Rate:** Affects the residence time and liquid holdup in the column, influencing performance.

Physical Consistency Checks

Ensuring Physical Plausibility

To validate the reliability of the surrogate models, their predictions were assessed for physical consistency within the expected operating domain.

Purity Domain Visualization

- The mole fractions of benzene in both the distillate () and bottoms () must lie within the theoretical range of $[0, 1]$. Visualizations confirm that the predicted values generally adhere to this fundamental physical constraint.

Model Stability Analysis

- **XGBoost Model:** The analysis revealed that the XGBoost model produced no violations for (always predicted within $[0, 1]$). For , one prediction was found to be slightly below zero, indicating a minor extrapolation issue or sensitivity at the extreme lower bound of the operating domain. This is a common challenge when models are pushed to their limits and highlights the importance of defining the valid prediction domain.
- **Other Models:** Similar checks were performed for Linear Regression, Random Forest, and ANN, with varying degrees of adherence to physical constraints. The XGBoost model demonstrated the best overall physical consistency among the explored options.

Final Model Selection

XGBoost: The Preferred Surrogate

Based on comprehensive evaluation, the **XGBoost model** was selected as the final surrogate for the benzene–toluene binary distillation column.

Key Performance Indicators (XGBoost)

- $x_D R^2$: 0.9043
- $x_B R^2$: 0.9447
- $QC R^2$: 0.9888
- $QR R^2$: 0.9173
- Average R^2 : 0.939

Why XGBoost?

Decision Graphic: Why XGBoost

This graphic summarizes the key reasons for selecting XGBoost, emphasizing its superior predictive accuracy (high R^2 values), robustness demonstrated through cross-validation, and efficiency in handling complex, non-linear relationships inherent in distillation processes.

Limitations and Considerations

Important Caveats

Warning Box:

Do not extrapolate beyond the validated DWSIM operating domain. The performance and accuracy of the surrogate model are guaranteed only within the range of conditions used to generate the training dataset. Extrapolation outside this domain may lead to unreliable predictions.

Dataset Specific Limitations

- **Stage Count:** The dataset is primarily based on distillation columns with either 10 or 12 stages. Performance for columns with significantly different stage counts may not be accurately captured.
- **Feed Flow Rate:** All simulations were conducted with a fixed feed molar flow rate of 10.00057 mol/s. The model's behavior at different feed flow rates has not been explicitly validated.
- **Component Specificity:** This surrogate model is specifically trained for the Benzene–Toluene binary system using the Peng–Robinson property package. Its applicability to other binary mixtures or different thermodynamic models is not assured.

Conclusion and Reproducibility

Summary of Findings

This project successfully developed a high-performance machine learning surrogate model for a benzene–toluene binary distillation column. By leveraging 147 unique operating points generated through rigorous DWSIM simulations, the XGBoost model was identified as the optimal solution, achieving an average R^2 of 0.939. This surrogate model offers a computationally efficient method for predicting key performance indicators such as product compositions (,) and energy duties (,).

Statement: *DWSIM provides the physics-based ground truth for complex process behavior, while machine learning provides a rapid surrogate prediction capability, enabling faster analysis and optimization.*

Reproducibility Package

To ensure the reproducibility of this work, the following components are provided:

- **Jupyter Notebook:** Contains all code for data preprocessing, model training, evaluation, and visualization.
- **CSV Dataset:** The complete dataset of 147 unique simulation results.
- **DWSIM Flowsheet:** The configured DWSIM simulation file for the distillation column.
- **Results Files:** Saved models and performance metrics for further analysis.

These materials facilitate verification and further development of the surrogate modeling approach.